

**13**

**B**

By Newton's 2<sup>nd</sup> law,

$$F_{\text{net}} \text{ on piston} = ma$$

Since gas exerts an upward force  $F_{\text{net}} = F_{\text{gas}} - mg$  where  $mg$  is the piston's weight

Hence

$$F_{\text{net}} = F_{\text{gas}} - mg = ma$$

$$F_{\text{gas}} = m(g + a) = 0.75 (9.81 + 3.5) = 9.983 \text{ N}$$

$$\text{Work done by the gas} = F_{\text{gas}} \times s = 9.983 \times 0.12 = 1.20 \text{ J}$$